Scientific Calculator

Mini Project Report

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1 Project Overview

1.1 Introduction

This report documents the development and deployment pipeline of a **Scientific Calculator** application built in Java. The project demonstrates a complete software development lifecycle including continuous integration, automated testing, containerization, and infrastructure-as-code deployment strategies.

Repository: `https://github.com/rahul09123/SPE-Mini-Project`

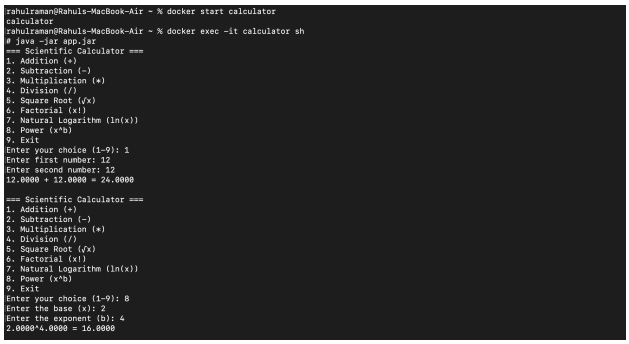

Figure 1: Scientific Calculator Application

1.2 Tech Stack & Objectives

Stack: Java 17, Maven 3, JUnit 5, Jenkins, Docker, Ansible

Application: Command-line scientific calculator with basic arithmetic (add, subtract, multiply, divide) and advanced functions (square root, factorial, logarithm, power) with comprehensive error handling.

2 Jenkins Pipeline Architecture

2.1 Pipeline Overview

The Jenkins pipeline automates the complete build, test, and deployment workflow. It follows a declarative pipeline approach, providing clarity and maintainability.

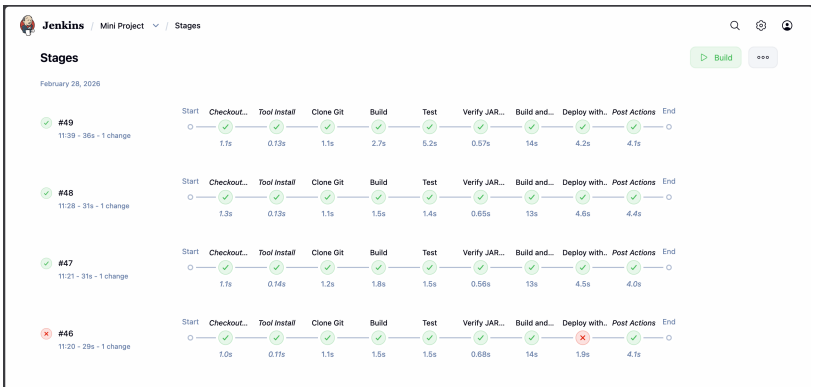

Figure 2: Jenkins Pipeline Architecture

2.2 ngrok: Exposing Jenkins to GitHub Webhooks

Since Jenkins runs locally on a development machine, it is not directly reachable from the public internet. **ngrok** is used to create a secure public tunnel to the local Jenkins server, enabling GitHub to send webhook events that automatically trigger pipeline builds on every code push.

2.2.1 How It Works

1. Jenkins is started locally and listens on `http://localhost:8080`.
2. ngrok is launched to forward a public HTTPS URL (e.g., `https://unfended-unexhibited-jude.ng` to that local port.
3. The generated ngrok URL is registered as a **GitHub Webhook** under the repository's *Settings* → *Webhooks* with the payload URL set to `https://unfended-unexhibited-jude.ngrok-`
4. Whenever a commit is pushed to the `main` branch, GitHub sends an HTTP POST to that URL.
5. ngrok forwards the request to the local Jenkins instance, which triggers the declarative pipeline automatically.

2.2.2 Key Benefits in This Project

- **No public server required:** Allows full CI/CD automation from a local machine without needing a cloud-hosted Jenkins instance.
- **Instant feedback loop:** Every `git push` triggers the complete pipeline (build → test → containerize → deploy) within seconds.
- **Secure tunnel:** ngrok provides TLS-encrypted HTTPS endpoints, satisfying GitHub's webhook security requirements.
- **Zero configuration overhead:** A single `ngrok http 8080` command establishes the tunnel, keeping the setup lightweight.

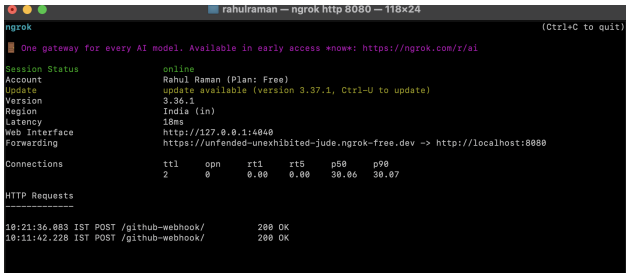

Figure 3: ngrok Tunnel Exposing Local Jenkins to the Internet

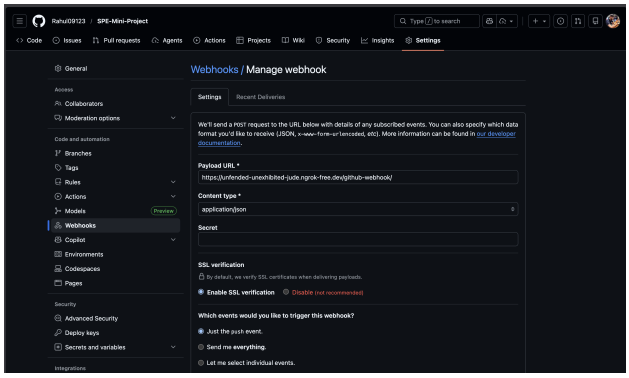

Figure 4: GitHub Webhook Configured with ngrok URL

2.3 Pipeline Stages

2.3.1 Stage 1: Clone Git

```
stage('Clone Git') {
    steps {
        script {
            git branch: 'main',
                credentialsId: 'github_credentials',
                url: "${GITHUB_REPO_URL}"
        }
    }
}
```

Listing 1: Clone Git Stage

Purpose: Fetches latest source code from GitHub. Authenticates to repository and provides clean workspace for builds.

2.3.2 Stage 2: Build

```
stage('Build') {
    steps {
        sh 'mvn clean package -DskipTests'
    }
}
```

Listing 2: Build Stage

Purpose: Compiles Java source and creates JAR artifact. Tests are skipped here and executed in the Test stage.

2.3.3 Stage 3: Test

```
stage('Test') {
    steps {
        sh 'mvn test'
    }
}
```

Listing 3: Test Stage

Purpose: Executes 47 comprehensive unit tests using JUnit 5 and generates test reports.

2.3.4 Stage 4: Verify JAR Existence

```
stage('Verify JAR Existence') {
    steps {
        sh 'ls -lh target/'
    }
}
```

Listing 4: Verify JAR Existence Stage

Purpose: Verifies successful JAR creation by listing target directory contents.

2.3.5 Stage 5: Build and Push Docker Image

```
stage('Build and Push Docker Image') {
    steps {
        script {
            sh """
            docker buildx create --use || true
            docker buildx inspect --bootstrap

            docker buildx build \
              --platform linux/amd64,linux/arm64 \
              -t ${DOCKER_HUB_USERNAME}/${DOCKER_IMAGE_NAME}:latest \
              --push .
            """
        }
    }
}
```

Listing 5: Docker Build Stage (Partial)

Purpose: Builds and pushes multi-platform Docker images (linux/amd64, linux/arm64) to Docker Hub registry.

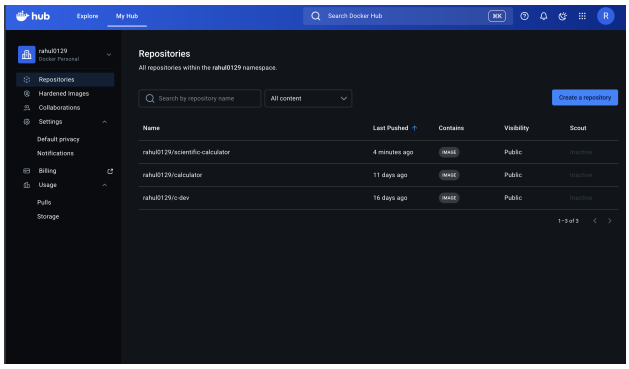

Figure 5: Docker Hub Multi-Platform Build

2.3.6 Stage 6: Deploy with Ansible

```
stage('Deploy with Ansible') {
    steps {
        sh 'ansible-playbook -i inventory.ini deploy.yml'
    }
}
```

Listing 6: Ansible Deployment Stage

Purpose: Automates application deployment to target servers using Ansible playbooks.

Figure 6: Container Running and Deployed

2.4 Environment Variables

```
environment {
    DOCKER_IMAGE_NAME = 'scientific-calculator'
    GITHUB_REPO_URL = 'https://github.com/rahul09123/SPE-Mini-Project.git'
    DOCKER_HUB_USERNAME = 'rahul0129'
    DOCKER_HOST = "unix:///Users/rahulraman/.docker/run/docker.sock"
}
```

Listing 7: Environment Configuration

These variables configure: Docker image name, GitHub repository URL, Docker Hub credentials, and Docker daemon socket.

3 Testing and Validation

3.1 Test Suite Overview

The project includes 47 comprehensive unit tests covering all calculator operations. Tests are organized using JUnit 5's annotation-based approach.

3.2 Test Coverage

3.2.1 Mathematical Operations Tested

- **Square Root:** Positive numbers, zero, decimals, negative numbers
- **Factorial:** Zero, one, positive integers (up to 20)
- **Natural Logarithm:** One, Euler's number, positive values, zero, negative values
- **Power Operations:** Base cases, decimal exponents, edge cases

3.2.2 Test Sample

Tests validate: Square Root ($\sqrt{16} = 4$, $\sqrt{0} = 0$), Factorial ($0! = 1$, $5! = 120$, $20! = 2.4\times10^{18}$), Natural Logarithm ($\ln(1) = 0$, $\ln(e) = 1$). Edge cases include NaN handling for negative numbers and infinity for $\ln(0)$. All 47 tests execute in 41 milliseconds with 100% pass rate.

4 Deployment Configuration

4.1 Dockerfile

The `Dockerfile` defines the container image used to package and run the Scientific Calculator application.

```
FROM openjdk:17.0.1-jdk-slim

WORKDIR /app

COPY target/SPEMiniProject-1.0-SNAPSHOT.jar app.jar

CMD ["/bin/sh"]
```

Listing 8: Dockerfile

4.1.1 Dockerfile Breakdown

- **Base Image (`FROM openjdk:17.0.1-jdk-slim`):** Uses a lightweight OpenJDK 17 slim image, providing the Java runtime while keeping the image size minimal.
- **Working Directory (`WORKDIR /app`):** Sets `/app` as the working directory inside the container, ensuring all subsequent operations occur in a predictable path.
- **Copy JAR (`COPY`):** Copies the Maven-built JAR artifact `SPEMiniProject-1.0-SNAPSHOT.jar` into the container and renames it to `app.jar` for simplicity.
- **Entry Point (`CMD ["/bin/sh"]`):** Opens an interactive shell on container start, allowing manual execution of the calculator via `java -jar app.jar` inside the container.

This image is built for multiple platforms (`linux/amd64` and `linux/arm64`) using Docker Buildx in the Jenkins pipeline and published to Docker Hub under `rahul0129/scientific-calculator:l`

4.2 Ansible Deployment Playbook (deploy.yml)

The `deploy.yml` file automates container deployment to target servers using Ansible orchestration.

4.2.1 Playbook Structure

```
- name: Pull the latest Docker image
  shell: docker pull rahul0129/scientific-calculator:latest

- name: Stop and remove existing container
  shell: docker rm -f calculator
  ignore_errors: true

- name: Run calculator container (detached)
  shell: docker run -dit --name calculator \
    --restart unless-stopped \
    rahul0129/scientific-calculator:latest

- name: Display container status
  shell: docker ps -f "name=calculator"
```

Listing 9: deploy.yml Tasks

4.2.2 Key Operations

- **Pull Image:** Downloads latest Docker image from Docker Hub
- **Cleanup:** Removes existing container (non-blocking with `ignore_errors`)
- **Deploy:** Launches container in detached mode with auto-restart policy
- **Verify:** Confirms successful container deployment and status

This playbook targets `webservers` host group defined in `inventory.ini`, enabling centralized deployment across multiple servers.

4.3 Pipeline Execution Flow

The Jenkins pipeline executes sequentially through all 6 stages: Clone (fetch code) → Build (Maven compile) → Test (JUnit 5, 47 tests) → Verify (JAR check) → Containerize (Docker multi-platform) → Deploy (Ansible).

4.4 Quality Assurance & Benefits

QA Measures: Automated testing with 47 test cases, build verification, multi-platform testing, and Infrastructure-as-Code deployment ensure reliability.

Benefits: Complete automation, consistency across environments, scalability to multiple servers, high portability across CPU architectures, and rapid feedback (41ms test execution time).

5 Conclusion

This project successfully demonstrates a complete CI/CD pipeline implementation with 100% test success (47/47 passing), fast execution (41ms), and multi-platform Docker containerization. The automated pipeline integrates Git-based version control, Maven-based Java builds, comprehensive JUnit 5 testing, multi-platform Docker image building and pushing, and Ansible-based infrastructure deployment. This comprehensive approach showcases modern DevOps practices combining continuous integration, automated testing, and infrastructure-as-code principles.

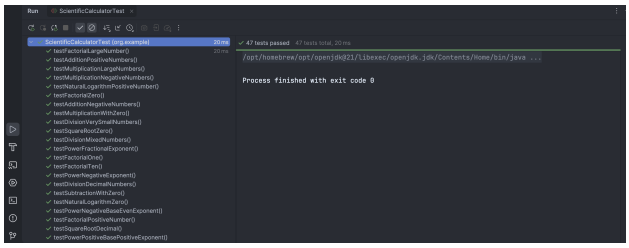

Figure 7: JUnit Test Results

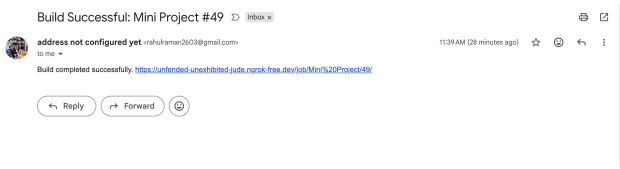

Figure 8: Email Notification on Build Status